Lesson 06: Designing a Study, Analyzing Data and Reporting Results

English Stage 4 (Psychology) · Unit 1: Psychology as a Science

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 2.6 Designing a Research Study; 2.7 Analyzing the Data; 2.8 Drawing Conclusions and Reporting the Results; 2.9 Overview of the Scientific Method (Summary).

In this lesson

You will follow the rest of the research process: variables, samples, designs, data analysis, conclusions and reports. On the language side, you will practice talking about evidence without "prove" and contrasting designs. Then you will write the design outline of your proposal.

Learning goals

After this lesson you should be able to:

- Identify the variables, operational definitions, population and sample of a study.
- Distinguish experimental from non-experimental, and laboratory from field, research in terms of validity.
- Explain descriptive and inferential statistics, statistical significance, and Type I and Type II errors.
- Explain why theories can be supported but not proved.
- Outline the design of your own study.

Key language

Term or phrase	Meaning	Example
variable	a quantity or quality that varies across people or situations	height; chosen major
quantitative / categorical variable	a quantity, measured with a number / a quality, measured with a category label	number of siblings / nationality
operational definition	a definition of a variable or construct in terms of precisely how it will be measured	depression as a score on the Beck Depression Inventory
population / sample	the large group researchers want to draw conclusions about / the subset of individuals actually studied	all university students / 20 student volunteers
independent variable	the variable the experimenter manipulates (the presumed cause)	bright or dim lighting
dependent variable	a variable thought to be the effect of the independent variable; it is measured	workers' productivity

Term or phrase	Meaning	Example
confounding variable	an extraneous variable that differs across the levels of the independent variable	noise that occurs only with bright lights
internal / external validity	how far the design supports a causal conclusion / how far results can be generalized beyond the people and situations studied	high in laboratory experiments / high in field studies
descriptive / inferential statistics	techniques for summarizing data / for drawing conclusions about a population from a sample	a mean / a test of significance
statistically significant	describes a result for which the null hypothesis (that there is no real effect) has been rejected: unlikely to be due to chance alone	a difference between conditions that is unlikely to be chance
Type I / Type II error	a false positive / failing to detect a real effect (a "missed opportunity")	reporting a nap effect that is not real / missing one that is
It is important to point out that ...	introduces a point readers might miss	"... non-experimental does not mean nonscientific."
The bottom line here is that ...	introduces the main conclusion	"... there is no such thing as scientific proof, there is only scientific evidence."

Rules and explanations

The ideas

Variables and samples (2.6). Most variables of interest, such as depression, cannot be observed directly, so researchers define them **operationally**: a score on the Beck Depression Inventory, the number of depressive symptoms, or a diagnosis of major depressive disorder. Researchers study a **sample** to draw conclusions about a **population**, so the sample should be representative. **Simple random sampling** (every member has an equal chance of selection) is rarely possible in psychology, so **convenience sampling** (people who are nearby and willing) is common, although it may not be representative.

Experimental versus non-experimental research

	Experimental	Non-experimental
Researchers ...	manipulate an independent variable, control extraneous variables, measure the dependent variable	measure variables as they naturally occur
Goals served	explain (causal conclusions)	describe and predict
Example	lighting (bright or dim) and workers' productivity	recording drivers' gender and phone use at an intersection

If the bright lights are noisy, noise is a **confound**: it varies with the independent variable, so we cannot tell whether light or noise caused a difference. If there is noise in both conditions, it is merely

an extraneous variable. (The book first defines extraneous variables as "any variable other than the dependent variable"; its lighting example shows it means any variable other than the independent and dependent variables.)

Laboratory versus field. Laboratory experiments control everything except the independent variable, so internal validity is high, but external validity is lower. Field studies, in real-world settings, have less control but higher external validity: there is a trade-off. **Field experiments** manipulate an independent variable in a natural setting and can have both.

Analyzing the data (2.7)

- **Descriptive statistics:** percentages; central tendency (mode: the most frequent score; median: the midpoint; mean: the average); dispersion (range; standard deviation, the average distance from the mean; variance, the standard deviation squared); and the correlation coefficient, from –1.00 to +1.00 (height and weight: positive; stressful events and happiness: negative). Experiments usually compare the means of the conditions.
- **Inferential statistics:** by convention, a result is statistically significant if it has less than a 5% chance of being due to random error. (This is the book's simplified wording; Unit 4 explains exactly what the 5% refers to.)
- **Errors:** lowering the threshold (to 1%, say) reduces Type I errors but increases Type II errors, which are also more likely with small samples.

Drawing conclusions (2.8). A significant result consistent with the hypothesis **supports** the theory; a disconfirmed hypothesis **weakens** it. Confirmation never proves a theory, and one disconfirmation does not simply disprove it; the theory may instead be modified (had Zajonc found no effect in cockroaches, he could have limited drive theory to animals with sufficiently complex nervous systems). A theory that cannot be modified to account for repeated disconfirmations must eventually be abandoned.

Reporting the results. The most prestigious way is an APA-style article in a peer-reviewed journal; others are chapters in edited books and conference talks or posters, which bring feedback before peer review.

Language focus

1. Evidence without "prove." "Scientists tend to avoid the word 'prove' when talking and writing about theories." Instead:

- results agree: the theory is **supported**; confirming a hypothesis can **strengthen** a theory; results are **consistent with** the hypothesis;
- results disagree: the hypothesis is **disconfirmed**; the theory **has been weakened** (or **refuted**);
- change: the evidence can be used to **modify** a theory; researchers may **abandon** it.

2. Contrasting designs.

- "A laboratory study is a study that is conducted in the laboratory environment. **In contrast**, a field study is a study that is conducted in the real-world ..."
- "**When** internal validity is high, external validity **tends to be** low"

- "However, **unlike** with experimental research, we cannot make causal conclusions using this method"

> **Note for Arabic speakers** - In academic writing, *data* is usually plural: "data **are** analyzed", "these data".

Practice

1. Reading questions

1. Why do researchers need operational definitions? Give two of the book's definitions of depression. (2.6)
2. Why is simple random sampling difficult in most psychological research, and what is the problem with convenience sampling?
3. Which goals of science can non-experimental research serve, and which can it not?
4. Why don't researchers simply set the significance threshold lower than 5%? (2.7)
5. Give three reasons why a disconfirmed hypothesis does not necessarily mean that a theory is wrong. (2.8)

2. Book exercises (Section 2.9, the exercises for Chapter 2. Without a university library, find open access articles with Google Scholar or in *Frontiers in Psychology*.)

1. Practice: Find a description of an empirical study in a professional journal or in one of the scientific psychology blogs. Then write a brief description of the research in terms of the cyclical model presented here. One or two sentences for each part of the cycle should suffice.
2. Practice: Generate three research ideas based on each of the following: informal observations, practical problems, and topics discussed in recent issues of professional journals.
3. Practice: Generate an empirical research question about each of the following behaviors or psychological characteristics: long-distance running, getting tattooed, social anxiety, bullying, and memory for early childhood events.
4. Practice: Evaluate each of the research questions you generated in Exercise 2 [Exercise 3 here] in terms of its interestingness based on the criteria discussed in this section.
5. Practice: Find a recent empirical research report in a professional journal. Read the introduction and highlight in different colors descriptions of theories and hypotheses.
6. Practice: Using the research article you found in a professional journal identify whether the study was experimental or non-experimental. If it was experimental identify the independent and dependent variables.
7. Practice: Using the research article you found in a professional journal identify which descriptive statistics were reported.
8. Practice: Describe why theories can be supported but not proved.

(Two other 2.9 exercises, finding 10 articles and book chapters and rating three studies for replication, need library access.)

3. Language exercise

A. Correct: (1) "Zajonc's cockroach study proved drive theory." (2) "The result was not statistically significant, so the researchers proved that there is no effect."

B. Join with *in contrast*: "Experimental research allows causal conclusions. Non-experimental research does not."

Speak with your tutor

- "Give me five short study descriptions. I will say whether each is experimental and name the independent and dependent variables."
- "PSYC 110 Lecture 18 (Mental Illness, Part I) describes patients with depression who received cognitive behavioral therapy, the drug Paxil, or a placebo. Ask me to name the variables and to explain what a Type I error would mean there."
- "PSYC 110 Lecture 17 says sleep is hard to study and describes EEG recordings in a sleep lab. Discuss whether an EEG would be feasible for my study."
- "Here is my design outline: [outline]. Act as a reviewer and find possible confounds."

Writing task

Research proposal, part 3: design outline. Write 300 to 400 words outlining a study to test your hypothesis from Lesson 05, under these headings: Variables; Operational definitions; Population and sample; Type of research; Setting and validity; Extraneous variables and confounds; Planned analysis; Reporting. Use the future tense ("Participants will ...") and evidence language (*support*, not *prove*).

Example notes for the default topic (sleep and memory in university students): independent variable, a nap opportunity of up to 90 minutes after learning, or resting quietly awake for the same time (categorical: nap or wake); dependent variable, the number of word pairs recalled out of 40 (quantitative); record how many minutes each nap participant actually slept; a convenience sample of 20 student volunteers, assigned at random to the nap or the wake group (a sample this small makes a Type II error more likely); an experiment, because the goal is to explain (a correlational alternative, usual hours of sleep and memory scores, could only describe and predict); a quiet laboratory room (high internal validity, but a lab nap is not like sleep at home); both groups learn the same list at the same time of day, and the awake group must not rehearse it; the mean and standard deviation of recall in each group, then a test of significance; a poster, then an APA-style article.

If you work on your own topic, use the same headings.

Self-editing checklist:

- Is each variable named, classified (quantitative or categorical) and operationally defined?
- Did I describe the sample and its limits?
- Did I justify the type of research by its goal (describe, predict or explain)?
- Did I name at least two extraneous variables and how I will control them?
- Did I discuss internal and external validity, and plan descriptive and inferential statistics?

- Did I use the future tense and "data are", and avoid "prove"?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "Not all scientific studies are created equal" (TED-Ed, David H. Schwartz), from the Chapter 2 exercises: https://youtu.be/GUpd2HJHUt8
- "How To Read an Academic Paper" (UBC iSchool), from the Chapter 2 exercises: https://youtu.be/SKxm2HF_-ko

Answer key

1. Reading questions

1. Most variables cannot be observed directly, but science depends on observation, so they must be turned into something observable and measurable. Depression: a score on the Beck Depression Inventory, the number of depressive symptoms, or a diagnosis of major depressive disorder (2.6).
2. Psychological populations (all teenagers, all children with autism) are less clearly defined than a list of a city's voters, so not every member can be given an equal chance. A convenience sample may not represent the population, so generalizing is less appropriate (2.6).
3. It can describe and predict; it cannot explain, because it does not allow causal conclusions (2.6).
4. Reducing the chance of a Type I error increases the chance of a Type II error (2.7, last paragraph).
5. It may be a Type II error; the design may be faulty (the independent variable not successfully manipulated or the dependent variable not well measured); or a minor, unstated assumption of the theory may not have been met (2.8).

2. Book exercises (example answers; many answers are possible)

1. Using the study in 2.1. *Question*: laptop note taking is common and suspected of harming learning, but earlier studies focused on distraction. *Study*: three studies compared laptop and longhand note takers. *Analysis*: laptop note takers did worse on conceptual questions. *Conclusion*: transcribing lectures verbatim is detrimental to learning. *Publication*: *Psychological Science* (2014), now part of the literature.
2. Observations: why some students always sit at the front; whether people eat more with friends; why people check their phones when they feel awkward. Problems: exam anxiety; patients who forget their medication; litter on campus. Journal topics (the 2.3 list): Do bilingual students learn a third language faster? Why do some people enjoy sad films? How accurate are police lineups?
3. Is weekly running distance related to anxiety? Do people with tattoos score higher on sensation seeking? How common is social anxiety among first-year students? Are bullied children more likely to report symptoms of depression? Do women and men differ in the age of their earliest memory?
4. Running: reasons for both answers (exercise may calm, heavy training may stress) and health implications: interesting. Tattoos: reasons for both answers: interesting. Social anxiety: interesting if not yet studied in this group. Bullying: only one answer seems likely, so refine it (online versus face-to-face bullying). Early memories: probably studied already (see 2.3), so check the literature.
5. to 7. Answers depend on your article. Theories often appear with *theory* or *model*, hypotheses with *predicted* or *hypothesized*. A study is experimental if the researchers manipulated a variable: the cell phone studies in 2.2 compared driving with and without a phone (independent variable: phone use; dependent variables: hazard detection, reaction time and vehicle control). Look for percentages, means, medians, standard deviations and correlations.
8. A confirming result may be a Type I error; other plausible theories may imply the same hypothesis, so confirming it supports them all; and a later test may always disconfirm it (the problem of

induction: no number of white swans proves that all swans are white). There is only evidence, not proof (2.8).

3. Language exercise

A. (1) Zajonc's cockroach study supported drive theory. (2) The result was not statistically significant, so the researchers could not conclude that there is an effect; a real effect may have been missed (a Type II error).

B. Experimental research allows causal conclusions; in contrast, non-experimental research does not.

My notes